Deep Learning for Economics and Finance

Foundations, boundary conditions, applications to ODEs and economic PDEs

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Based on: Raissi, Perdikaris & Karniadakis (2019), *J. Comput. Physics* 378, 686–707.

Sirignano & Spiliopoulos (2018), Deep Galerkin Method for PDEs in high dimensions.

Al-Aradi, Correia, Naez, Hoshisashi & Schwab (2018), DGM PDE solver.

Materials: course GitHub repository

Outline

1. **From DEQNs to PINNs**
   Bridging discrete and continuous time; the PINN loss.
2. **PINN Fundamentals**
   Automatic differentiation for PDEs, collocation, training loop.
3. **Soft vs. Hard Boundary Conditions**
   Trial solutions, 2D Poisson, practical guidance.
4. **Economic Applications: HJB Equation**
   Cake-eating problem, scaled trial solution + hard BCs, Adam $\rightarrow$ L-BFGS, analytical benchmark; DGM as optional high-D architecture.
5. **Financial Applications: Black–Scholes**
   Option pricing via PINNs, comparison with analytical formula.

Roadmap of This Lecture

From DEQNs to PINNs

- Discrete vs. continuous time
- The PINN loss function
- DEQN vs. PINN comparison

PINN Fundamentals

- Autograd for PDEs
- Collocation and training
- Warm-up: 1D ODE example

Boundary Conditions

- Soft vs. hard enforcement
- Trial solutions
- 2D Poisson example

Econ & Finance

- HJB / cake-eating: trial solution + hard BCs
- Adam warm-start, L-BFGS polish, FP64
- Black–Scholes PDE; DGM as optional architecture

Part I

From DEQNs to PINNs

Recall: The DEQN Loss from Day 1

Yesterday: we trained a neural network $\mathcal{N}_\theta$ to satisfy economic equilibrium conditions in discrete time:

$$\boxed{\ell_\theta^{\text{DEQN}} = \frac{1}{N}\sum_{i=1}^{N}\left\| G\left(\boldsymbol{x}_i, \mathcal{N}_\theta(\boldsymbol{x}_i)\right)\right\|^2}$$

- $G(\cdot) = 0$: Euler equations, budget constraints, market clearing
- Training points $\boldsymbol{x}_i$: simulated from the model's own dynamics
- No labeled data needed—structural equations *are* the loss

Today's question: What if the economic model is formulated in continuous time?

Discrete vs. Continuous Time

Discrete Time (DEQN)

Euler equation:

$$u'(c_t) = \beta\,\mathbb{E}[u'(c_{t+1})\,R_{t+1}]$$

State transition:

$$\boldsymbol{x}_{t+1} = h(\boldsymbol{x}_t, p(\boldsymbol{x}_t), \varepsilon_{t+1})$$

Loss: algebraic residual $G(\cdot) = 0$

Continuous Time (PINN)

HJB equation:

$$\rho\,V = \max_c \{u(c) + V_a\,\dot{a}\}$$

State evolution:

$$\mathrm{d}a = (ra - c)\,\mathrm{d}t + \sigma\,\mathrm{d}W$$

Loss: PDE residual $\mathcal{R}(\cdot) = 0$

Key shift: from algebraic residuals to **differential** residuals—we need derivatives of the neural network.

Why Continuous Time Matters

Many frontier models are formulated in continuous time:

- Heterogeneous-agent models: Achdou, Han, Lasry, Lions & Moll (2022)—income and wealth distribution dynamics via HJB–KFE systems
- Mean-field games: optimal policy with a continuum of interacting agents
- Asset pricing: Black–Scholes, Heston, jump-diffusion models
- Macro-finance: Brunnermeier & Sannikov (2014), He & Krishnamurthy (2013)

Central bank relevance

Continuous-time models are the workhorse for financial stability analysis, option pricing, and monetary transmission in modern central banking.

The PINN Loss Function

Core idea (Raissi, Perdikaris & Karniadakis, 2019):

Given a PDE $\mathcal{D}[u](\boldsymbol{x}) = 0$ on domain Ω with boundary conditions $\mathcal{B}[u](\boldsymbol{x}) = 0$ on $\partial\Omega$, define:

$$\boxed{\ell_\theta^{\mathrm{PINN}} = \underbrace{\frac{1}{N_r}\sum_{i=1}^{N_r}\left|\mathcal{D}[\mathcal{N}_\theta](\boldsymbol{x}_i^r)\right|^2}_{\text{PDE residual}} + \underbrace{\frac{\lambda}{N_b}\sum_{j=1}^{N_b}\left|\mathcal{B}[\mathcal{N}_\theta](\boldsymbol{x}_j^b)\right|^2}_{\text{boundary/initial conditions}}}$$

- $\mathcal{N}_\theta$: neural network with parameters θ
- $\boldsymbol{x}_i^r$: collocation points in the interior of Ω
- $\boldsymbol{x}_j^b$: points on the boundary $\partial\Omega$
- Derivatives $\mathcal{D}[\mathcal{N}_\theta]$ computed via automatic differentiation

Warm-up ODE: for $u''(x) = -1$ on $(0, 1)$ with $u(0) = u(1) = 0$, use $\mathcal{D}[u] = u'' + 1$ in the interior and $\mathcal{B}[u] = (u(0), u(1))$ on the boundary.

PAUSE — 5 min Finger Exercise: Writing a PINN Loss

Exercise

Consider the ODE $u''(x) = -1$ on $[0, 1]$ with boundary conditions $u(0) = u(1) = 0$.

1. Write down the PINN residual $\mathcal{R}(x; \theta) = \ldots$ using a neural network $\mathcal{N}(x; \theta)$.
2. Write the total PINN loss as a sum of a PDE residual term and a boundary term.
3. What is the exact solution?

Solution: Writing a PINN Loss

Answer

1. $\mathcal{R}(x;\theta) = \frac{\partial^2 \mathcal{N}}{\partial x^2}(x;\theta) + 1.$
2. $\mathcal{L}(\theta) = \underbrace{\frac{1}{N_r}\sum_{i=1}^{N_r}\left|\mathcal{R}(x_i;\theta)\right|^2}_{\text{PDE residual}} + \lambda \underbrace{\left[\mathcal{N}(0;\theta)^2 + \mathcal{N}(1;\theta)^2\right]}_{\text{boundary conditions}}$
3. $u(x) = \frac{1}{2}x(1-x)$ satisfies $u'' = -1,\ u(0) = u(1) = 0.$ ✓

DEQN vs. PINN: Side-by-Side

DEQN (Discrete Time)

Economic eqns $G(\cdot) = 0$

$\downarrow$

$\mathcal{N}_\theta(\boldsymbol{x}_t) \approx p(\boldsymbol{x}_t)$

$\downarrow$

$\ell = \|G(\boldsymbol{x}, \mathcal{N}_\theta)\|^2$

$\downarrow$

SGD $\rightarrow \theta^\star$

same principle

$\longleftrightarrow$

+ autograd derivatives

PINN (Continuous Time)

PDE $\mathcal{D}[u] = 0$ + BCs

$\downarrow$

$\mathcal{N}_\theta(\boldsymbol{x}) \approx u(\boldsymbol{x})$

$\downarrow$

$\ell = |\mathcal{D}[\mathcal{N}_\theta]|^2 + \lambda|\mathrm{BC}|^2$

$\downarrow$

SGD $\rightarrow \theta^\star$

Both approaches are unsupervised: the physics (economics) *is* the loss function. The key addition for PINNs: we differentiate the network via `torch.autograd.grad`.

Part II

PINN Fundamentals

Automatic Differentiation for PDEs

Key tool: `torch.autograd.grad` with `create_graph=True`.

```
# First derivative dy/dx
dydx = torch.autograd.grad(
    y, x,
    grad_outputs=torch.ones_like(y),
    create_graph=True # keep graph for higher derivs
)[0]
```

- `create_graph=True`: retains the computational graph so we can differentiate again
- This gives us algorithmic derivatives (up to floating-point precision)—no finite differences!
- Works for any order: $\frac{\partial^n}{\partial x^n}$ by chaining n calls

Computing Second Derivatives

Chain two autograd calls to get $y''(x)$:

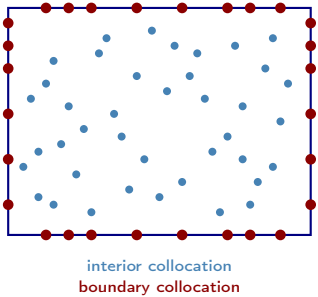
```
# First derivative
dydx = torch.autograd.grad(
    y, x, grad_outputs=torch.ones_like(y),
    create_graph=True)[0]

# Second derivative
d2ydx2 = torch.autograd.grad(
    dydx, x, grad_outputs=torch.ones_like(dydx),
    create_graph=True)[0]
```

Collocation Points

PINNs evaluate the PDE residual at randomly sampled points—no grid required.

1D: interior + boundary

2D: $\Omega = (0, 1)^2$



Points are resampled periodically during training to avoid overfitting to a fixed set.

PINN Training Algorithm

Algorithm: Physics-Informed Neural Network Training

Input: PDE $\mathcal{D}[u] = 0$, BCs $\mathcal{B}[u] = 0$, network $\mathcal{N}_\theta$, learning rate η
for epoch $= 1, \ldots, E$ **do**
  Sample N_r collocation points $\{\boldsymbol{x}_i^r\}$ in domain Ω
  Sample N_b boundary points $\{\boldsymbol{x}_j^b\}$ on $\partial\Omega$
  Forward pass: compute $\mathcal{N}_\theta(\boldsymbol{x}_i^r)$ and $\mathcal{N}_\theta(\boldsymbol{x}_j^b)$
  Autograd: compute $\mathcal{D}[\mathcal{N}_\theta](\boldsymbol{x}_i^r)$ via `torch.autograd.grad`
  $\ell_\theta = \frac{1}{N_r}\sum_i |\mathcal{D}[\mathcal{N}_\theta](\boldsymbol{x}_i^r)|^2 + \frac{\lambda}{N_b}\sum_j |\mathcal{B}[\mathcal{N}_\theta](\boldsymbol{x}_j^b)|^2$
  Update: $\theta \leftarrow \theta - \eta\,\nabla_\theta \ell_\theta$
end for
Output: Trained $\mathcal{N}_{\theta^\star}$ approximating the PDE solution

Note: Unlike DEQNs, training points are not simulated from model dynamics—they are drawn uniformly (or adaptively) from the domain.

Optimizer Pipeline: Adam $\rightarrow$ L-BFGS, in FP64

Two stages.

- Adam on resampled collocation: finds a basin.
- Deterministic L-BFGS on a *fixed* grid in FP64: closes the residual; resampling per eval breaks the line search.

Why FP64?

- tanh-MLP 2nd derivatives lose $\sim$7 digits in FP32.
- L-BFGS Wolfe search compares values at exactly that scale.
- FP32 plateaus at the noise floor; FP64 keeps shrinking.

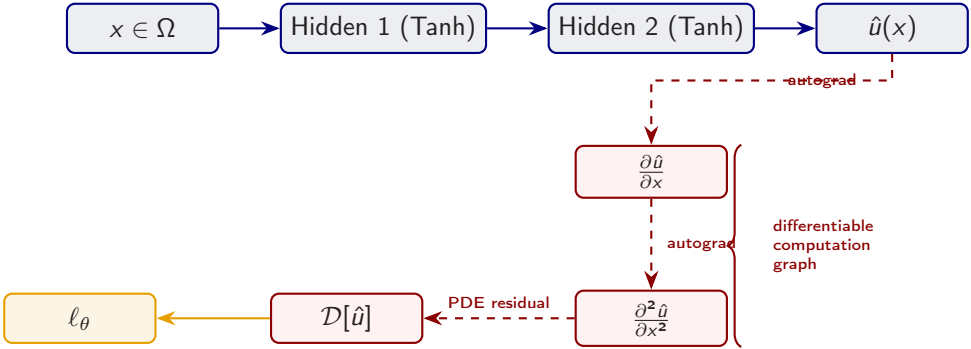
Practical recipe

1. Cast model + data to `float64`.
2. Adam, lr 10^{-3}, a few k epochs, resampled collocation.
3. Freeze grid; `torch.optim.LBFGS` (strong-Wolfe).
4. Stop on residual stagnation.

Recent precision evidence: arXiv:2505.10949.

Used in notebooks 04, 06, 07, 08: L-BFGS is the deterministic polish step. Final residuals remain problem-dependent, so report PDE residuals and solution diagnostics rather than relying on the optimizer label.

PINN Architecture

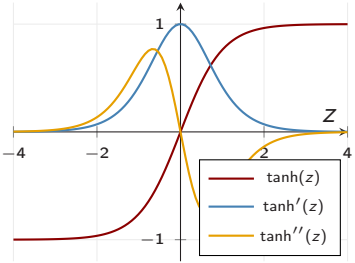


The entire pipeline—forward pass *and* derivative computation—is differentiable, enabling end-to-end gradient-based training.

Why Tanh Activations for PINNs?

PDE residuals require smooth derivatives.

- ReLU: C^0 only—second derivative is zero everywhere (useless for u'')
- Tanh: C^∞—all derivatives are well-defined and informative
- Softplus: C^∞, useful for positivity constraints (e.g. consumption)



Rule of thumb

For PINNs requiring k-th order derivatives, use activations that are at least C^k.

Warm-up: Solving an ODE with a PINN

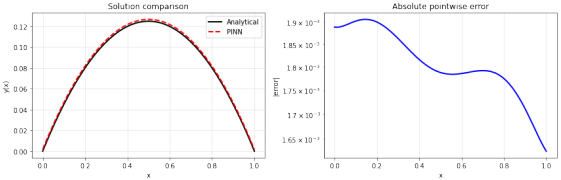
Problem: $y''(x) = -1$ on $(0,1)$, $\quad y(0) = y(1) = 0$. **Analytical:** $y(x) = \frac{1}{2}x(1-x)$.

PINN setup:

1. NN $\mathcal{N}_\theta : [0,1] \to \mathbb{R}$, 2 layers, 20 units
2. Residual: $r(x) = \mathcal{N}_\theta''(x) + 1$
3. Loss: PDE residual + BC penalty

$$\ell = \frac{1}{N}\sum_i r(x_i)^2 + \mathcal{N}_\theta(0)^2 + \mathcal{N}_\theta(1)^2$$

Notebook: `01_ODE_PINN_ZeroBCs`



Trained PINN matches analytical solution; pointwise error $\sim 10^{-3}$.

Non-Zero Boundary Conditions: The Challenge

Problem: $y''(x) = -1$ on $(0, 1)$ with $y(0) = 1$, $y(1) = 2$.

Analytical solution: $y(x) = -\frac{1}{2}x^2 + \frac{3}{2}x + 1$.

Soft BC enforcement:

$$\ell = \frac{1}{N}\sum_i r(x_i)^2 + \lambda_0\big(\mathcal{N}_\theta(0) - 1\big)^2 + \lambda_1\big(\mathcal{N}_\theta(1) - 2\big)^2$$

Issues with soft enforcement:

- BCs are only approximately satisfied
- Penalty weight λ requires tuning: too small $\Rightarrow$ BC violated; too large $\Rightarrow$ PDE ignored
- Boundary errors can propagate into the interior

$\Rightarrow$ **Motivates hard boundary enforcement** (Part III).

Part III

Soft vs. Hard Boundary Conditions

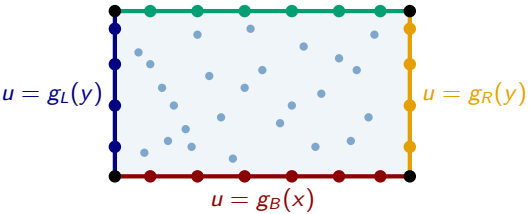
What Are Boundary Conditions?

1D: $y''(x) = -1$ **on** $(0, 1)$

Interior: enforce $y'' + 1 = 0$

Boundary: enforce y = prescribed value

2D: $\Delta u = f$ **on** $(0, 1)^2$

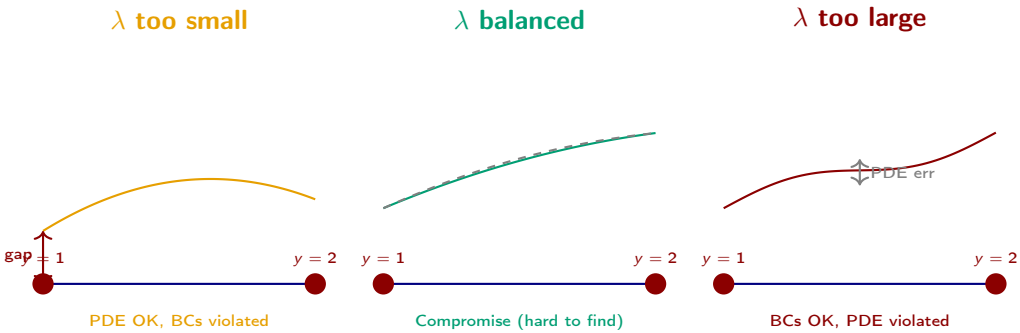


Two enforcement strategies

Soft: add BC violation as a penalty term in the loss.

Hard: build BCs into the network output by construction.

The Problem with Soft Boundary Conditions



Soft BCs require careful balancing of the penalty weight λ.
Too small: BCs violated. Too large: PDE ignored. Just right: hard to find *a priori*.

Hard BC Enforcement: Trial Solutions

Idea: construct a trial solution that satisfies BCs *by construction*:

$$\boxed{\hat{y}(x) = A(x) + B(x) \cdot \mathcal{N}_\theta(x)}$$

where:

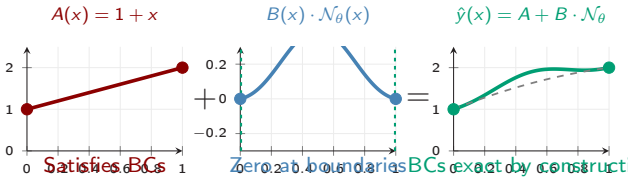
- $A(x)$: anchor function satisfying the BCs exactly, e.g. $A(x) = 1 + x$
- $B(x)$: mask function vanishing at boundaries, e.g. $B(x) = x(1 - x)$
- $\mathcal{N}_\theta(x)$: free neural network (no BC constraints)

Verification:

$$\hat{y}(0) = A(0) + B(0) \cdot \mathcal{N}_\theta(0) = 1 + 0 = 1 \quad \checkmark$$
$$\hat{y}(1) = A(1) + B(1) \cdot \mathcal{N}_\theta(1) = 2 + 0 = 2 \quad \checkmark$$

$\Rightarrow$ BCs hold identically—only the PDE residual loss remains.

Trial Solution: Visual Decomposition



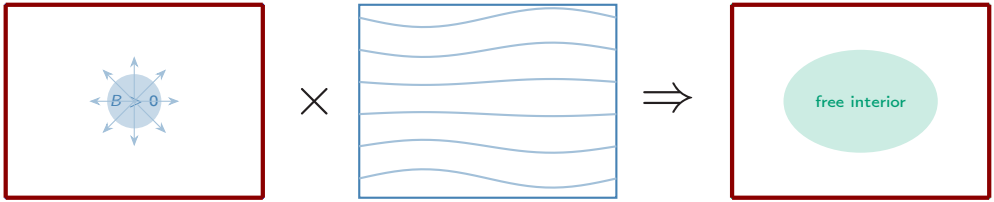
Hard BC: Implementation

```
def A(x): # Anchor: A(0)=1, A(1)=2
    return 1.0 + x
def B(x): # Mask: B(0)=B(1)=0
    return x * (1.0 - x)

def y_trial(model, x): # Hard-BC trial solution
    return A(x) + B(x) * model(x)

def residual(model, x):
    x.requires_grad_(True)
    y = y_trial(model, x)
    dydx = torch.autograd.grad(y, x, ...)[0]
    d2ydx2 = torch.autograd.grad(dydx, x, ...)[0]
    return d2ydx2 + 1.0 # should be 0
```

Advantage: the loss has only one term (PDE residual)—no λ to tune!

Hard BCs in 2D: Building the Trial Solution

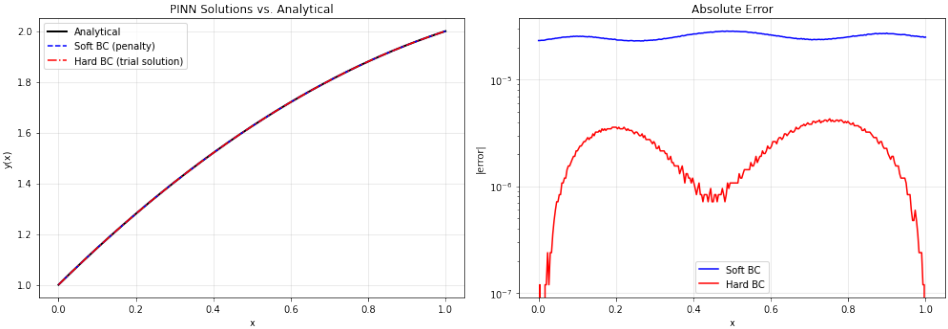
Mask $B(x,y) = x(1-x)\,y(1-y)$ **Network** $\mathcal{N}_\theta(x,y)$ $B \cdot \mathcal{N}_\theta$**: zero on** $\partial\Omega$



$B = 0$ on all edges Free to take any value Exactly zero on boundary

Then $\hat{u}(x,y) = \underbrace{A(x,y)}_{\text{matches BCs}} + \underbrace{B(x,y)\cdot\mathcal{N}_\theta(x,y)}_{\text{zero on } \partial\Omega}$ satisfies Dirichlet BCs exactly.

Soft vs. Hard BCs: Comparison



Hard BCs eliminate boundary error by construction, so the optimizer only fits the PDE interior. The notebook compares this against the tuned soft-penalty baseline.

Notebook: `02_ODE_PINN_SoftVsHardBCs`

PAUSE — 5 min Finger Exercise: Hard Boundary Conditions

Exercise

Consider the same ODE $u''(x) = -1$ on $[0, 1]$ with $u(0) = u(1) = 0$.

1. Construct a hard-BC trial solution of the form $\hat{u}(x) = B(x) \cdot \mathcal{N}(x; \theta)$, where $B(x)$ enforces both BCs exactly. (Hint: B must vanish at $x = 0$ and $x = 1$.)
2. Verify that $\hat{u}(0) = \hat{u}(1) = 0$ for *any* network output $\mathcal{N}$.
3. What is the advantage of hard BCs over soft BCs?

Solution: Hard Boundary Conditions

Answer

1. $\hat{u}(x) = x(1-x) \cdot \mathcal{N}(x;\theta)$. Since $B(x) = x(1-x)$ vanishes at both endpoints, the BCs are satisfied exactly.
2. $\hat{u}(0) = 0 \cdot \mathcal{N}(0;\theta) = 0. \quad \hat{u}(1) = 0 \cdot \mathcal{N}(1;\theta) = 0.$ ✓
   This holds regardless of what $\mathcal{N}$ outputs.
3. Hard BCs eliminate the boundary loss term entirely—the optimizer only needs to minimize the PDE residual. This reduces the number of hyperparameters (λ is gone) and often speeds up convergence.

Extension to 2D: Poisson Equation

Problem: $\Delta u = f(x, y)$ on $\Omega = (0, 1)^2$ with Dirichlet BCs.

Manufactured solution: $u^\star(x, y) = x^2 + y + \sin(\pi x)\sin(\pi y)$.

Hard BCs via transfinite (Coons-patch) interpolation — the standard 4-side construction that matches each boundary function exactly along its edge:

$$\hat{u}(x, y) = A(x, y) + \underbrace{x(1-x)\,y(1-y)}_{B(x,y)} \cdot \mathcal{N}_\theta(x, y)$$

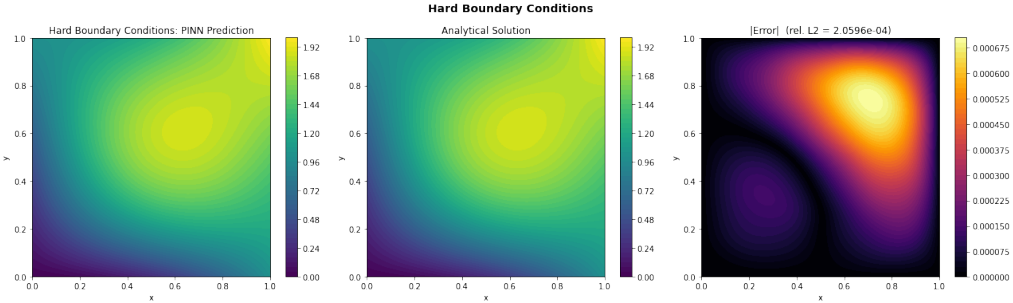
where $A(x, y)$ interpolates boundary data from all four sides:

$$A = (1-x)\,g_L(y) + x\,g_R(y) + (1-y)\,g_B(x) + y\,g_T(x) - \text{corners}$$

Laplacian via autograd:

$$\Delta \hat{u} = \frac{\partial^2 \hat{u}}{\partial x^2} + \frac{\partial^2 \hat{u}}{\partial y^2} \quad \text{(two chains of } \texttt{autograd.grad}\text{)}$$

2D Poisson: Results



Hard-BC PINN, analytical solution, and absolute error.
Relative L^2 error $\approx 2.06 \times 10^{-4}$. Boundary conditions satisfied exactly.

Notebook: `03_PDE_PINN_Poisson2D`

Soft vs. Hard BCs: When to Use Which?

	Soft BCs	**Hard BCs**
Best for	Complex geometries, time-dependent BCs, Robin/Neumann	Simple domains, Dirichlet BCs, high accuracy
Pros	Flexible, easy to implement	Exact BCs, fewer hyperparams, better accuracy
Cons	λ tuning, approximate BCs	Needs analytical A, B; harder for irregular domains

Practical recommendation

Start with hard BCs when the domain and boundary data allow it. Fall back to soft BCs for complex geometries or mixed boundary types.

Part IV

Economic Applications: The HJB Equation

The Hamilton–Jacobi–Bellman Equation

Generic continuous-time optimization:

$$\rho\, V(\boldsymbol{x}) = \max_{c} \left\{ u(c) + V_{\boldsymbol{x}} \cdot f(\boldsymbol{x}, c) \right\}$$

- $V(\boldsymbol{x})$: value function (unknown)
- ρ: discount rate
- $u(c)$: flow utility
- $f(\boldsymbol{x}, c)$: law of motion for state $\boldsymbol{x}$
- $V_{\boldsymbol{x}} = \nabla V$: gradient of the value function

This is a PDE! First-order (or second-order with diffusion).

$\Rightarrow$ Perfect fit for PINNs: the network approximates V, autograd gives $V_{\boldsymbol{x}}$, and the HJB residual is the loss.

Cake-Eating Problem: Setup

A household chooses consumption $c(t)$ to maximize lifetime utility:

$$\max_{\{c(t)\}} \int_0^\infty e^{-\rho t} \frac{c(t)^{1-\gamma}}{1-\gamma} \, \mathrm{d}t \qquad \text{s.t.} \quad \dot{a} = r\,a - c$$

HJB equation: $\rho\, V(a) = \max_c \left\{ \frac{c^{1-\gamma}}{1-\gamma} + V'(a)\,(r\,a - c) \right\}$

FOC: $c^{-\gamma} = V'(a) \;\Rightarrow\; c^\star = \big(V'(a)\big)^{-1/\gamma}$.

Analytical solution (with $\kappa = \frac{\rho - (1-\gamma) r}{\gamma} > 0$):

$$V^\star(a) = \frac{\kappa^{-\gamma}}{1-\gamma}\, a^{1-\gamma}, \qquad c^\star(a) = \kappa\, a$$

Parameters: $\gamma = 2$, $\rho = 0.05$, $r = 0.03 \;\Rightarrow\; \kappa = 0.04$.

PINN for the HJB Equation: Hard-BC Trial Solution

Anchor + mask + scaled MLP. Let $x(a) = (a - a_{\min})/(a_{\max} - a_{\min}) \in [0, 1]$ and S a value-scale constant:

$$\hat{V}(a) \;=\; V^\star(a_{\min}) + x(a)\big[V^\star(a_{\max}) - V^\star(a_{\min})\big] \;+\; x(a)\big(1 - x(a)\big)\, S\, f_\theta\big(x(a)\big).$$

Endpoints are exact *by construction*, so the boundary loss is identically zero and the optimizer focuses on the HJB residual only.

Consumption via FOC, $\hat{c}(a) = (\hat{V}'(a))^{-1/\gamma}$, with softplus safeguard on $\hat{V}'$.

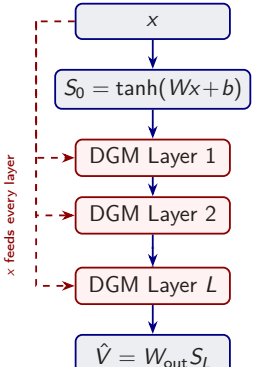
Loss: only the interior HJB residual remains,

$$\ell \;=\; \frac{1}{N}\sum_i \mathcal{R}(a_i)^2, \qquad \mathcal{R}(a) = \rho\hat{V} - \left[\tfrac{\hat{c}^{1-\gamma}}{1-\gamma} + \hat{V}'(ra - \hat{c})\right].$$

Pedagogical point: a small MLP plus hard BCs beats a large gated net with a soft-BC penalty, because the boundary term no longer dominates the objective.

Optional Architecture for High-D PDEs: DGM

Deep Galerkin Method (Sirignano & Spiliopoulos, 2018) — a heavier architecture that becomes worth its weight only in higher dimensions or with sharp PDE features. For the 1D cake problem, the trial-solution MLP on the previous slide is preferred.



DGM Layer gates (LSTM-like, Hochreiter & Schmidhuber 1997); x-skip connections recall Highway Networks (Srivastava et al. 2015). $S_\ell \in \mathbb{R}^U$ is the depth-ℓ hidden state ($S_0 = \tanh(Wx + b)$), $\odot$ is Hadamard, Z, G, R are sigmoid gates in $[0, 1]$:

$$Z_\ell = \sigma(U^z x + W^z S_\ell + b^z)$$
$$G_\ell = \sigma(U^g x + W^g S_\ell + b^g)$$
$$R_\ell = \sigma(U^r x + W^r S_\ell + b^r)$$
$$H_\ell = \tanh(U^h x + W^h (S_\ell \odot R_\ell) + b^h)$$
$$S_{\ell+1} = (1 - G_\ell) \odot H_\ell + Z_\ell \odot S_\ell$$

Key idea: feeding x into every layer helps the network resolve sharp PDE gradients.

DGM Implementation (PyTorch)

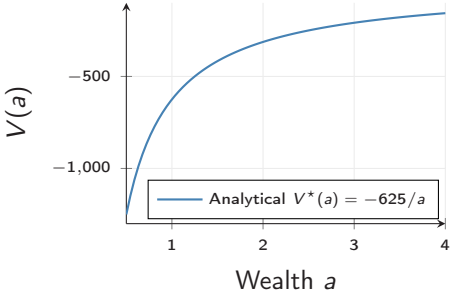
```python
class DGMNet(nn.Module):
    def __init__(self, dim, layers=4, units=64):
        super().__init__()
        self.input = nn.Linear(dim, units)
        self.z = nn.ModuleList(nn.Linear(dim, units) for _ in range(layers))
        self.wz = nn.ModuleList(nn.Linear(units, units) for _ in range(layers))
        # ... similar for g, r, h gates
        self.out = nn.Linear(units, 1)

    def forward(self, x):
        s = torch.tanh(self.input(x))
        for l in range(len(self.z)):
            z = torch.sigmoid(self.z[l](x) + self.wz[l](s))
            g = torch.sigmoid(self.g[l](x) + self.wg[l](s))
            r = torch.sigmoid(self.r[l](x) + self.wr[l](s))
            h = torch.tanh(self.h[l](x) + self.wh[l](s * r))
            s = (1 - g) * h + z * s
        return self.out(s)
```

HJB Residual Implementation

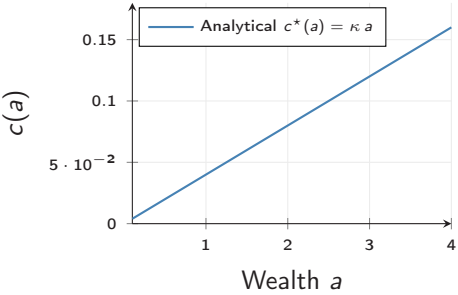
```python
def pde_residual(model, a):
    a.requires_grad_(True)
    V = model(a) # value function
    V_a = torch.autograd.grad( # V'(a) via autograd
        V.sum(), a, create_graph=True)[0].squeeze(-1)

    safe_Va = F.softplus(V_a) + 1e-6 # positivity guard
    c = safe_Va.pow(-1.0 / gamma) # FOC: optimal c
    u_c = c.pow(1-gamma) / (1-gamma) # flow utility

    R = rho*V - (u_c + safe_Va*(r*a.squeeze(-1) - c))
    return R # HJB residual
```

The `softplus` safeguard prevents $V'(a)^{-1/\gamma}$ from blowing up if $V'(a) \leq 0$ during early training.

Cake-Eating: Expected Behavior (Schematic)



Value function $V(a)$ (closed form)



Consumption policy $c(a)$ (closed form)

Validation target: closed form. Classroom run: final HJB loss $\approx 3 \times 10^{-4}$; consumption relative L^2 error $\approx 6 \times 10^{-3}$ after Adam + FP64 L-BFGS.

Notebook: `04_Cake_Eating_HJB_PINN`

Extensions: From Cake-Eating to Frontier Models

The PINN/DGM framework scales naturally:

1. **Income risk:** add diffusion $\mathrm{d}a = (ra - c + y)\,\mathrm{d}t + \sigma\,\mathrm{d}W$ $\Rightarrow$ second-order HJB
2. **Aiyagari-type models:** HJB + Kolmogorov Forward Equation $\Rightarrow$ coupled PDE system
3. **General equilibrium:** prices clear markets $\Rightarrow$ hybrid DEQN + PINN
4. **High dimensions:** multiple assets, heterogeneous agents $\Rightarrow$ no grid needed!

See: Achdou et al. (2022), Fernandez-Villaverde et al. (2024), Gopalakrishna (2024).

Part V

Financial Applications: Black–Scholes

The Black–Scholes PDE

European call option (S: asset, K: strike, T: maturity):

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

Terminal: $V(S,T) = \max(S-K,0)$

Boundary conditions:

- $V(0,t) = 0$
- $V(S_{\max},t) = S_{\max} - Ke^{-r(T-t)}$

PINN approach:

- Network $\mathcal{N}_\theta(S,t) \approx V(S,t)$
- Autograd: V_t, V_S, V_{SS}
- 4 loss terms:

ℓ_{PDE}	interior residual
ℓ_{BC_0}	$V(0,t) = 0$
ℓ_{term}	$V(S,T) = \text{payoff}$
$\ell_{\text{BC}_{S_{\max}}}$	far-field BC

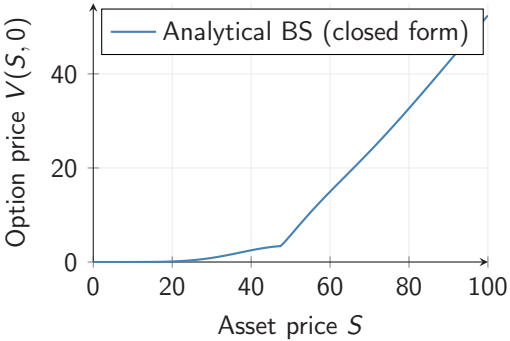
Black–Scholes PINN: Key Code

```
def pde_residual(model, S, t):
    S.requires_grad_(True)
    t.requires_grad_(True)
    X = torch.cat([S, t], dim=1)
    V = model(X)

    V_t = torch.autograd.grad(V, t, ...)[0]
    V_S = torch.autograd.grad(V, S, ...)[0]
    V_SS = torch.autograd.grad(V_S, S, ...)[0]

    # Black-Scholes PDE residual
    res = V_t + 0.5*sigma**2*S**2*V_SS + r*S*V_S - r*V
    return res
```

Loss:

$$\ell = \mathrm{MSE}(\mathrm{res}) + \mathrm{MSE}(V|_{S=0}) + \mathrm{MSE}(V|_{t=T} - \mathrm{payoff}) + \mathrm{MSE}(V|_{S_{\max}} - \mathrm{target})$$

Black–Scholes: Expected Behavior (Schematic)



Parameters: $K = 50$, $T = 1$, $r = 0.05$, $\sigma = 0.2$, $S_{\max} = 100$.

Network: 3 hidden layers, 50 units, tanh activations, 10,000 epochs (Adam).

Schematic of the call-option price curve. Current notebook diagnostics report max absolute error ≈ 0.13 and mean absolute error ≈ 0.04 against the closed form.

Notebook: `05_Black_Scholes_PINN`

Summary and Key Takeaways

Method

PINNs embed differential equations directly into the neural network loss function via automatic differentiation.

Boundary Conditions

Hard BCs (trial solutions) eliminate tuning and ensure exact satisfaction. Use soft BCs for complex domains.

Applications

HJB equations, Black–Scholes, and any continuous-time model. Scales to high dimensions without grids.

From DEQNs to PINNs:

- Same philosophy: economic/financial equations *are* the loss
- New ingredient: `torch.autograd.grad` for derivatives of $\mathcal{N}_\theta$
- Broader scope: any PDE, not just algebraic equilibrium conditions

Questions?

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https://sischei.github.io

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Companion Notebooks:

01 ODE PINN (Zero BCs)
02 Soft vs. Hard BCs
03 2D Poisson
04 Cake-Eating HJB
05 Black–Scholes

Next up (May 5th):

Continuous-Time Heterogeneous Agent Models: HJB + KFE